PharmAchieve

ULCERATIVE COLITIS AND CROHN’S DISEASE

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

Providing Care: Clinical Care

Providing Care: Distribution

Knowledge & Expertise

Communication & Collaboration

Leadership & Stewardship

Professionalism

LEGEND

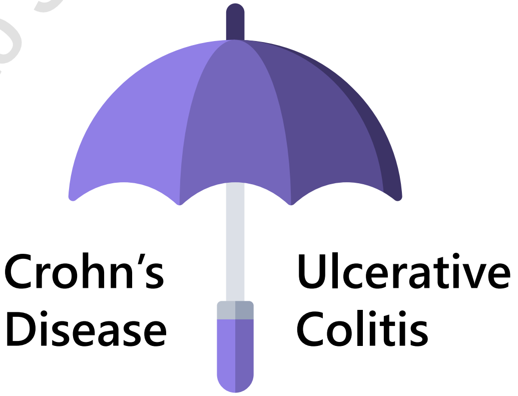
5-ASA	5-Aminosalicylic Acid
6-MP	6-Mercaptopurine
ACEi	Angiotensin Converting Enzyme Inhibitor
AZA	Azathioprine
BG	Blood Glucose
CD	Crohn’s Disease
CDAI	Crohn's Disease Activity Index
ESR	Erythrocyte Sedimentation Rate
GIT	Gastrointestinal Tract
IBD	Inflammatory Bowel Disease
MTX	Methotrexate
N/V	Nausea and Vomiting
NSAID	Non-Steroidal Anti-Inflammatory Drug
TB	Tuberculosis
TMP/SMX	Trimethoprim/Sulfamethoxazole
TMPT	Thiopurine Methyltransferase Polymorphism
UC	Ulcerative Colitis

PATHOPHYSIOLOGY

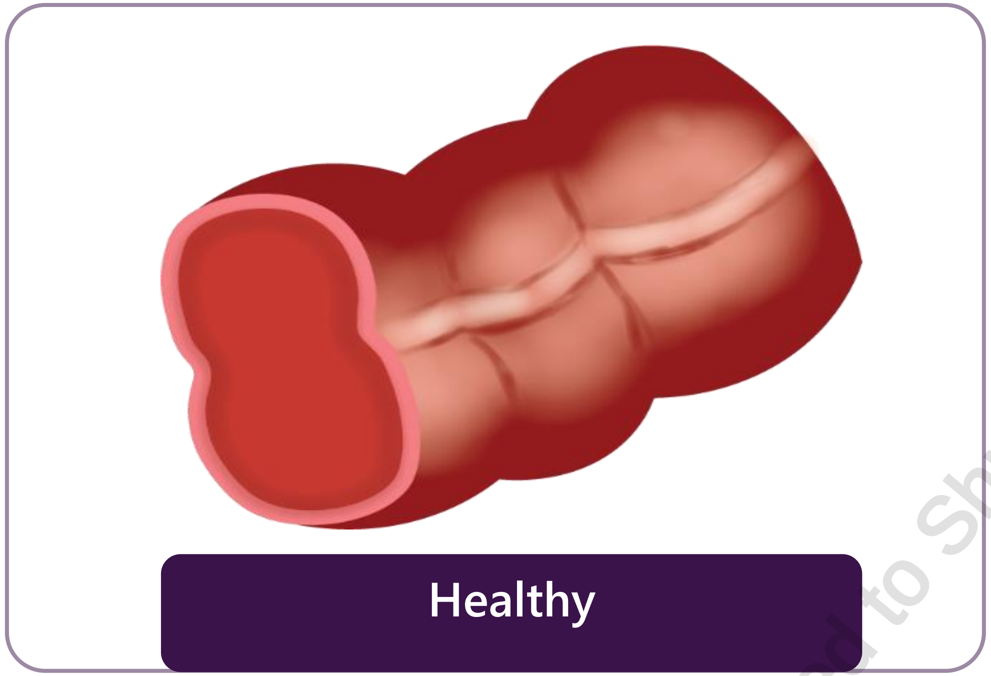
Inflammatory Bowel Disease (IBD)

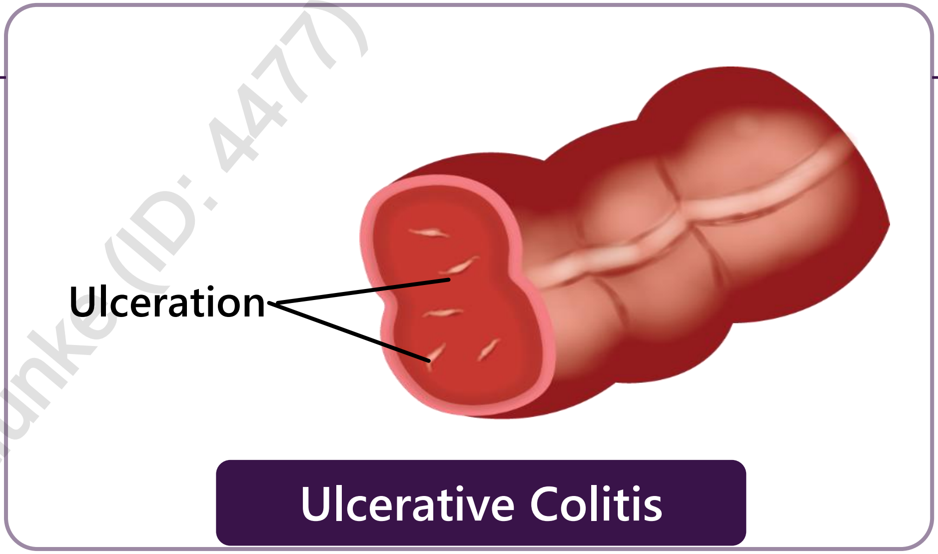
- Group of inflammatory disorders in the gastrointestinal tract
- Interferes with ability to digest food, absorb nutrients and eliminate waste
- Due to unknown cause, immune system mounts a response against healthy tissue in digestive system leading to inflammation
  - True cause of IBD is unclear

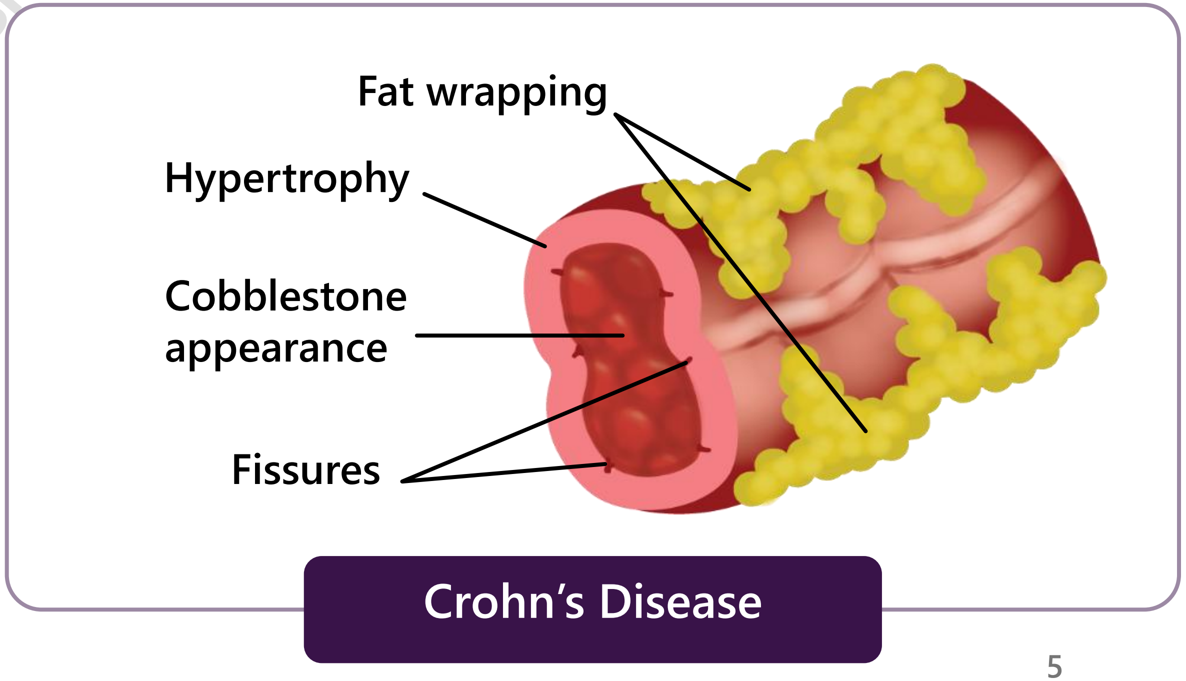
- Chronic disease with periods of active disease/flare and no symptoms/remission
- Treatments target immune system: 5-ASA, corticosteroids, immunosuppressants
- Moderate-to-severe IBD: biologics



CLINICAL PRESENTATION



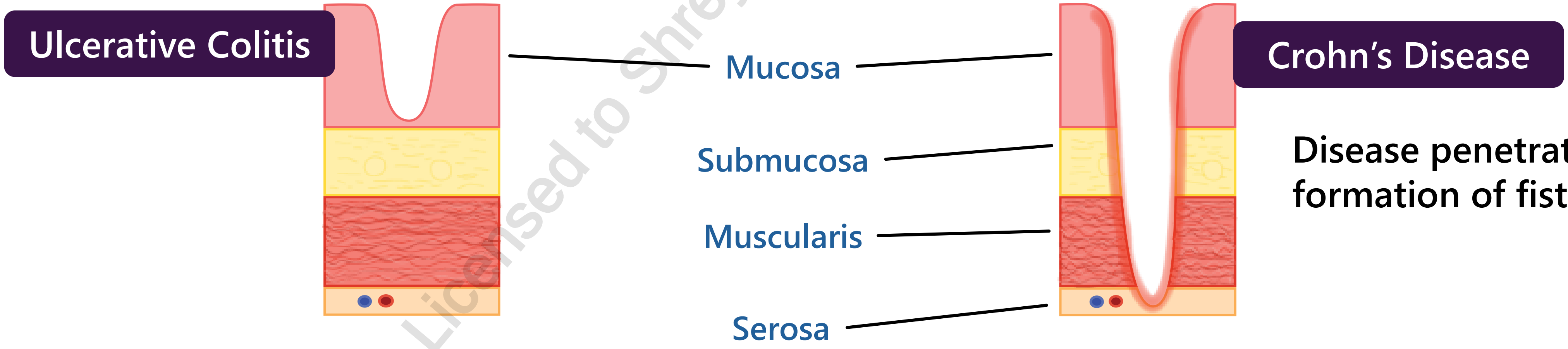




CLINICAL PRESENTATION: DISEASE EXTENT

	Ulcerative Colitis (UC)	Crohn's Disease (CD)
Extent	• Inflammation is continuous, diffuse, and shallow • Limited to mucosal layer	• Inflammation is discontinuous, patchy, and transmural • Entire thickness of bowel wall to the outermost layer of the digestive tract can be affected

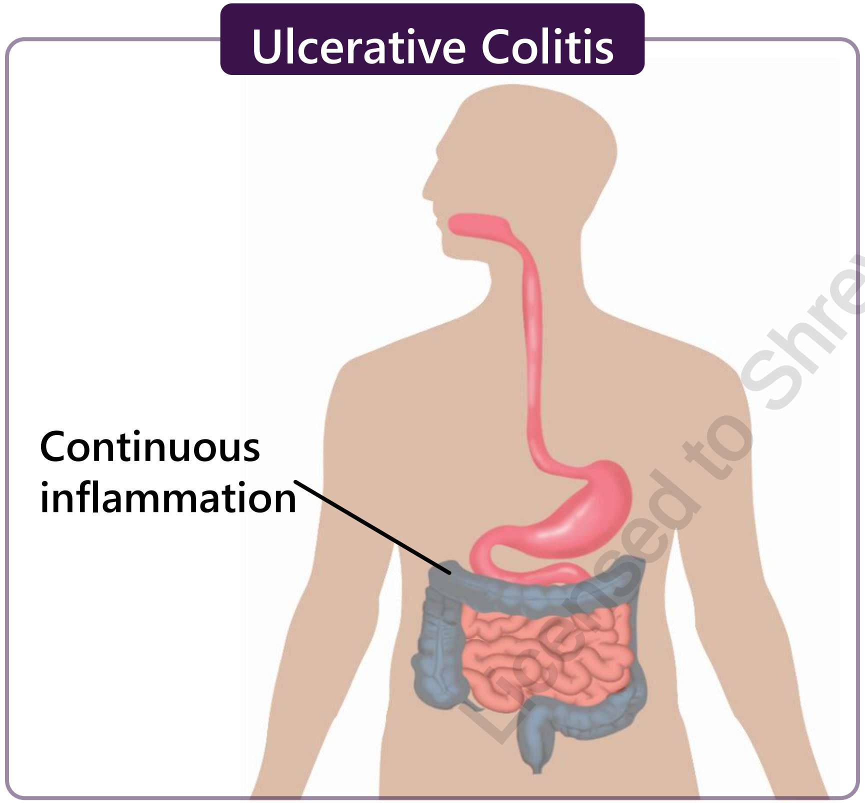
Layers of the intestinal wall

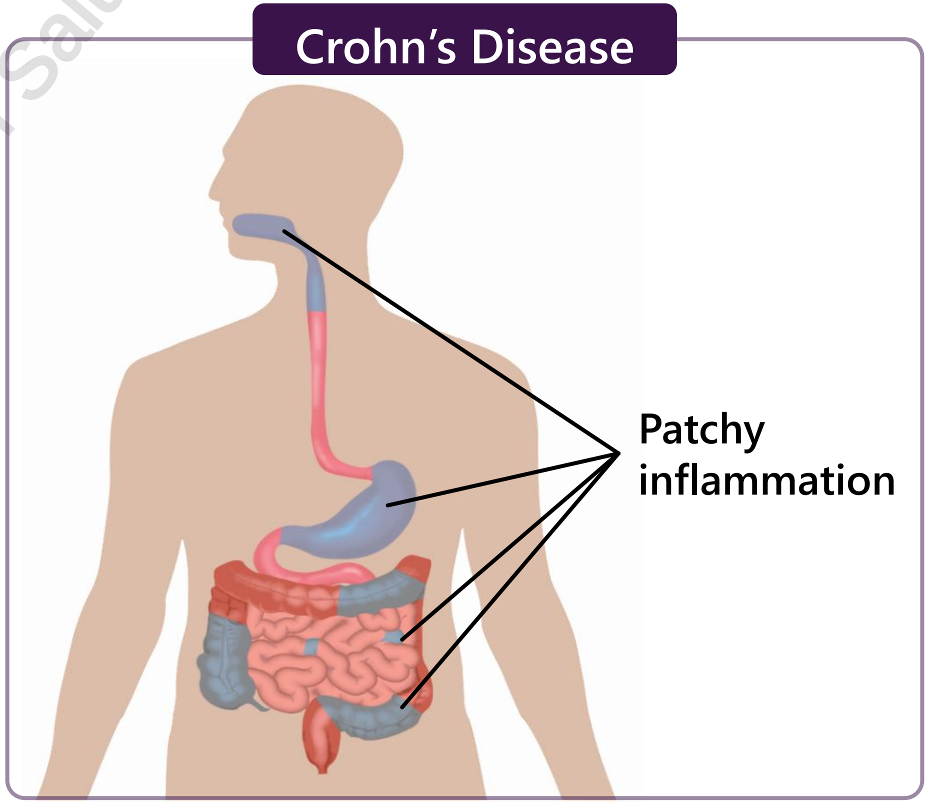


Disease penetration, formation of fistula

CLINICAL PRESENTATION: DISEASE LOCATION

	Ulcerative Colitis (UC)	Crohn’s Disease (CD)
Location	• Limited to colon and rectum • Usually starts at rectum and extends up in a continuous manner	• Can occur anywhere in GIT (mouth to anus) • Usually lower part of small bowel and upper colon affected and rectum is spared





CLINICAL PRESENTATION

	Ulcerative Colitis (UC)	Crohn's Disease (CD)
Unique symptoms	• Common: rectal bleeding, cryptic abscesses • May include fever, fissures, strictures and abdominal mass • Complications: bleeding from ulcerations, perforation of bowel	• Common: N/V, abdominal mass/pain, fissures, strictures, obstructions, sinus tracts, perforations, fistulas between organs (consider: multiple layers affected) • Rare: rectal bleeding, cryptic abscess, toxic megacolon • Complications: hemorrhage, fulminant colitis, toxic megacolon • TNFα is a pivotal pro-inflammatory cytokine involved
General symptoms	Abdominal pain/cramps, change in bowel habits, diarrhea, weight loss, decreased appetite, anemia, fatigue, extraintestinal manifestations, increased risk of colon cancer	

CLINICAL PRESENTATION- UNDERSTANDING SYMPTOMS

- **Severe diarrhea:** due to decreased colonic absorption of water/electrolytes
- **Abdominal pain and cramping:** due to spasms of irritated GI tract
- **Rectal bleeding:** due to mucosal erosions
- **Weight loss:** due to nutrient malabsorption from damage to small intestine
- **Extraintestinal manifestations:** arthritis, hepatobiliary disease, ocular (iritis, uveitis, etc.), skin lesions, thromboembolism, renal stones, osteoporosis

DIAGNOSIS

<u>Requires a clinical diagnosis that combines:</u>

1. Endoscopy and/or colonoscopy
   - Visualize and identify areas of inflammation in the GI tract
   - Biopsy allows for depth of involvement to be determined

2. Barium small bowel X-ray
   - Look for small bowel thickening/inflammation

3. Clinical History, physical exam, laboratory tests, CT/MRI
   1. Positive fecal calprotectin
   2. Elevated CRP/ESR
   3. Leukocytosis

4. Rule out differential diagnosis (e.g. infection, ischemia)

UC: Severity of disease (Mild, Moderate or severe) is based on stool frequency, rectal bleeding, endoscopic findings and overall patient assessment

CD: Severity of disease (Mild, Moderate or severe) is based on having multiple risk factors, the presence of deep ulcers on endoscopic appearance and overall disease burden and location

RISK FACTORS

- Family history (genetics)
- Age <30 years, may be a second peak between ages 50-80
- Environmental triggers:
  - bacterial/viral/enterobacterial infection/pollution
- Dietary triggers: inflammatory foods, western diet
- Smoking
  - UC: ***Smoking cessation*** increases risk of UC
  - CD: ***Smoking*** increases risk of CD
- Psychological stress
- Caucasian ethnicity (European origin) or South Asian
- Vitamin D deficiency

MEDICAL/DRUG RELATED CAUSES

- NSAIDs: may exacerbate symptoms but do not cause UC/CD
- Oral contraceptives: controversial – do not recommend stopping oral contraceptives in patients that are currently on it
- Use with caution:
  - Opioids, anticholinergics : reduce GI motility, may lead to narcotic bowel syndrome/risk of bowel obstruction and toxic megacolon
  - Antidiarrheals: risk of toxic megacolon in moderate to severe cases

GOALS OF THERAPY

- ✓ Induce and maintain steroid-free remission
- ✓ Relieve current signs and symptoms and improve quality of life
- ✓ Improve nutritional status and weight in children
- ✓ Reduce risk of colon cancer & complications
- ✓ Address extraintestinal manifestations

NON-PHARMACOLOGICAL MEASURES

- Surgery
  - UC: cures UC
  - CD: may be needed to treat acute complications (perforation, obstruction, abscesses, toxic megacolon), refractory disease; CD recurs after surgery
- Bowel rest
- Optimize individual diet and avoid food triggers
- Hydration
- Smoking cessation

PHARMACOLOGICAL MEASURES

Aminosalicylates (oral and rectal)	• 5-ASA/Mesalamine, Sulfasalazine
Corticosteroids (oral and rectal)	• Prednisone, Budesonide, Hydrocortisone, IV methylprednisolone
Immunosuppressants	• Thiopurines: Azathioprine, 6-mercaptopurine • Antimetabolite: Methotrexate
Biologic Therapies	
Anti-TNFα therapies	• Infliximab, Adalimumab, Golimumab
Anti-integrin therapies	• Vedolizumab
Anti-interleukin therapies	• Ustekinumab

PHARMACOLOGICAL MEASURES- AMINOSALICYLATES

Drugs:	Sulfasalazine, Mesalamine/5-ASA (Pentasa®, Salofalk®, Mezavant®) Olsalazine (Dipentum®)
MOA:	Topical anti-inflammatory effect
Efficacy:	UC: 1st line - all equally effective in inducing/maintaining remission for mild-moderate UC CD: sulfasalazine has modest benefit inducing remission in mild colonic CD
Interactions:	• Myelosuppression with azathioprine / 6-mercaptopurine • Renal failure with other nephrotoxins (e.g. NSAIDS)
AEs:	• Abdominal pain, cramps, diarrhea (↑olsalazine), headache, blood dyscrasias • Rare: pancreatitis • Sulfasalazine: increased ADR, rare idiosyncratic reports of renal impairment + risk hemolytic anemia/folate deficiency/ hepatotoxicity due to sulfapyridine moiety/caution in patients with allergy to sulfonamides or salicylates
Tips:	• Titrate up to minimize GI upset, abrupt discontinuation not recommended • Releases in the small bowel (ileum) to colon: Salofalk®, Pentasa® • Releases in the colon: sulfasalazine (colonic bacteria breaks down sulfasalazine into sulfapyridine and mesalamine), olsalazine, Mezavant® • Pentasa® and Salofalk® available as oral, suppository & rectal enema • Mezavant® is the only once-daily mesalamine • *No evidence that patients with a history of allergy to acetylsalicylic acid could not safely take 5-ASA preparations.*

PHARMACOLOGICAL MEASURES- CORTICOSTEROIDS

Drugs:	Budesonide (oral/rectal), Hydrocortisone (oral/IV), Prednisone (oral), Methylprednisolone (IV)
MOA:	↓ pro-inflammatory cytokine/mediator release, modulates immune system
Efficacy:	UC: 1st line for induction of remission in moderate-severe UC/CD (more effective than 5-ASA in both UC and CD) CD: Budesonide as 1st line in mild-moderate ileal and or right colon CD Not used for maintenance of remission
Interactions:	• NSAIDs (↑GI ulceration), antidiabetics (steroids may ↑BG) • Budesonide is extensively metabolized by CYP3A4 • Strong CYP3A4 inducers such as phenobarbital, phenytoin, and rifampin can ↑ clearance
AEs:	GI upset, dyspepsia, edema, hyperglycemia, hypokalemia, suppression of HPA axis, impaired wound healing, psychosis, Cushing syndrome features, osteoporosis, cataracts, myopathy (if on prolonged therapy), sleep and mood disturbance
Tips:	• Taper systemic corticosteroids by 5-10 mg weekly until 25-20 mg, then by 2.5 mg/weekly; tapering is required to prevent relapse and withdrawal symptoms from steroids • Offer osteoporosis prophylactic therapy with adequate doses of calcium and vitamin D • Monitoring: annual eye exam, bone mineral density, blood sugars, and monitor for infections • Budesonide CR (Entocort®) → for right-sided ileal disease only – high first-pass metabolism limits systemic effects • Budesonide MMX (Cortiment®) → for entire colon • Budesonide enema→ for distal colon

PHARMACOLOGICAL MEASURES – IMMUNOSUPPRESSANTS: THIOPURINES

Drugs:	Azathioprine (Imuran®), 6-mercaptopurine (Purinethol®) (active metabolite of azathioprine)
MOA:	Purine antagonist, inhibits DNA/RNA synthesis
Efficacy:	• UC/CD: Maintenance of remission for moderate-severe UC/CD, not for induction of remission as monotherapy • UC: Can be combined with anti-TNF therapy to augment their efficacy as both induction and maintenance • Delayed onset of action up to 3-4 months, up to 6 months for a therapeutic effect
Interactions:	Allopurinol and febuxostat (inhibits xanthine oxidase, which leads to ↑6-MP and increased risk of 6-MP toxicity, bone marrow suppression), immunosuppressants, myelosuppressive drugs (e.g. clozapine, TMP/SMX)
AEs:	N/V, fever, infection, rash, arthralgia, bone marrow suppression, ↑ risk of lymphoma, and nonmelanoma skin cancer, hepatotoxicity, blood dyscrasias, pancreatitis (rare)

Tips:	• Low risk in pregnancy and breastfeeding
	• Baseline LFTs and CBCs should be performed
	• Reduce dose in oliguric patients
	• Thiopurines are metabolized by TPMT which converts 6MP to inactive metabolites.
	• Thiopurine methyltransferase (TMPT) polymorphism: TPMT levels may be low or absent in some patients,
	    • If ↓TMPT, ↑6MP, ↑ bone marrow suppression: TPMT assay is necessary before initiation of treatment to identify patients at risk for severe dose dependent myelosuppression
	    • Monitor CBC weekly x 4, then q1-2 months
	• Patients should have yearly skin checks and be counselled on measures to reduce UV exposure, such as use of sun block and wearing sun-protective clothing

PHARMACOLOGICAL MEASURES- IMMUNOSUPPRESSANTS-ANTIMETABOLITE

Drugs:	Methotrexate (IM/SC)
MOA:	Folate antimetabolite – inhibits DNA synthesis
Efficacy:	CD: Induction of complete remission in moderate-severe steroid-resistant/dependent CD UC: efficacy not well-established - do not use MTX as monotherapy to induce or maintain remission) Onset: 3-6 weeks, up to 12 weeks
Interactions:	Alcohol, NSAIDs (↑MTX concentration), penicillins (↑MTX concentration)
AEs:	• Infection, nausea, flu-like aches, headaches, myelosuppression, hepatotoxicity • Rare: hepatitis, pneumonitis, increased risk of lymphoma
Tips:	• Do not use in pregnancy/breastfeeding; recommend effective contraception; stop 3-12 months prior to attempted conception • Co-administer with folic acid supplementation to minimize adverse drug reactions related to folate deficiency (e.g. alopecia and stomatitis) • Oral MTX not studied in controlled trials • Baseline LFTs, CBCs and pregnancy test should be done

PHARMACOLOGICAL MEASURES- BIOLOGICS- TNFA ANTAGONISTS

Drugs:	Infliximab IV (Remicade®), Adalimumab SC (Humira®), Golimumab SC (Simponi®)
MOA:	Monoclonal antibodies that neutralizes effect of TNFα (pro-inflammatory cytokine)
Efficacy:	UC and CD: 2/3 of patients show response, with 1/3 showing remission for both moderate-severe UC and CD (drugs act to induce and maintain remission) Used in patients who do not respond to thiopurines or corticosteroids. Recommended to start TNFa antagonists with either thiopurine or methotrexate to reduce secondary loss of response
Interactions:	Immunosuppressants, live vaccines
AEs:	• Infusion reactions, headache, rash, arthralgias, opportunistic infections, abdominal pain, reversible lupus-like syndrome, worsening heart failure, lymphoma, CNS-demyelinating disorders • Infliximab is associated with an increased risk of anti-antibody formation and anaphylaxis as it is a mouse/human chimeric antibody
Tips:	• Ensure immunizations are up-to-date • Screen for TB, hepatitis B/C and varicella prior to initiation of treatment

PHARMACOLOGICAL MEASURES- BIOLOGICS- INTEGRIN INHIBITOR

Drugs:	Vedolizumab IV (Entyvio®)
MOA:	Gut-selective anti-inflammatory monoclonal antibody that binds exclusively to $\alpha4\beta7$ integrin on lymphocytes and selectively inhibits adhesion to mucosal addressin cell adhesion molecule 1 (MAdCAM-1)
Efficacy:	Inducing and maintaining remission in patients with moderate-severe CD + UC, including patients who failed prior TNFα antagonist therapies, corticosteroids and thiopurines
Interactions:	Immunosuppressants, live vaccines
AEs:	Infections, nasopharyngitis, headache, arthralgia, nausea, pyrexia, upper respiratory tract infection, fatigue, cough, bronchitis, influenza, back pain, rash, pruritis, sinusitis, oropharyngeal pain, pain in extremities
Tips:	• Ensure immunizations are up-to-date • Screen for TB, hepatitis B/C and varicella prior to initiation of treatment

PHARMACOLOGICAL MEASURES- BIOLOGICS- INTERLEUKIN ANTAGONIST

Drugs:	Ustekinumab IV (Stelara®)
MOA:	Monoclonal antibody that binds to the common p40 protein subunit of IL-12 and IL-23 to prevent activation of inflammatory and immune responses
Efficacy:	Remission and induction in moderate-severe UC and CD with inadequate response to corticosteroids, immunomodulators, TNFα antagonists
Interactions:	Immunosuppressants, live vaccines
AEs:	Infections, arthralgia, headache, nasopharyngitis, injection site erythema, fatigue, vulvovaginal mycotic infections, diarrhea, pruritus, nausea
Tips:	• Ensure immunizations are up-to-date • Screen for TB, hepatitis B/C and varicella prior to initiation of treatment

TREATMENT ALGORITHM- MILD-MODERATE UC

<u>Left-sided colitis:</u>
Rectal 5-ASA ≥1 g/day (enema preferred over suppository) <u>OR</u>
Oral MMX budesonide <u>OR</u>
Oral 5-ASA 2-4.8 g/day ± rectal 5-ASA≥1 g /day

<u>Extensive colitis:</u>
Oral 5-ASA 2-4.8 g/day ± rectal 5-ASA ≥1 g/day <u>OR</u>
Oral MMX budesonide

4-8 weeks

Symptomatic response?
- Yes → Continue 5-ASA (oral or rectal)
- No → Optimize 5-ASA therapy †

4-12 months

Complete remission?
- Yes → Continue 5-ASA (oral or rectal)
- No → *2nd line:* Oral or rectal corticosteroids ‡

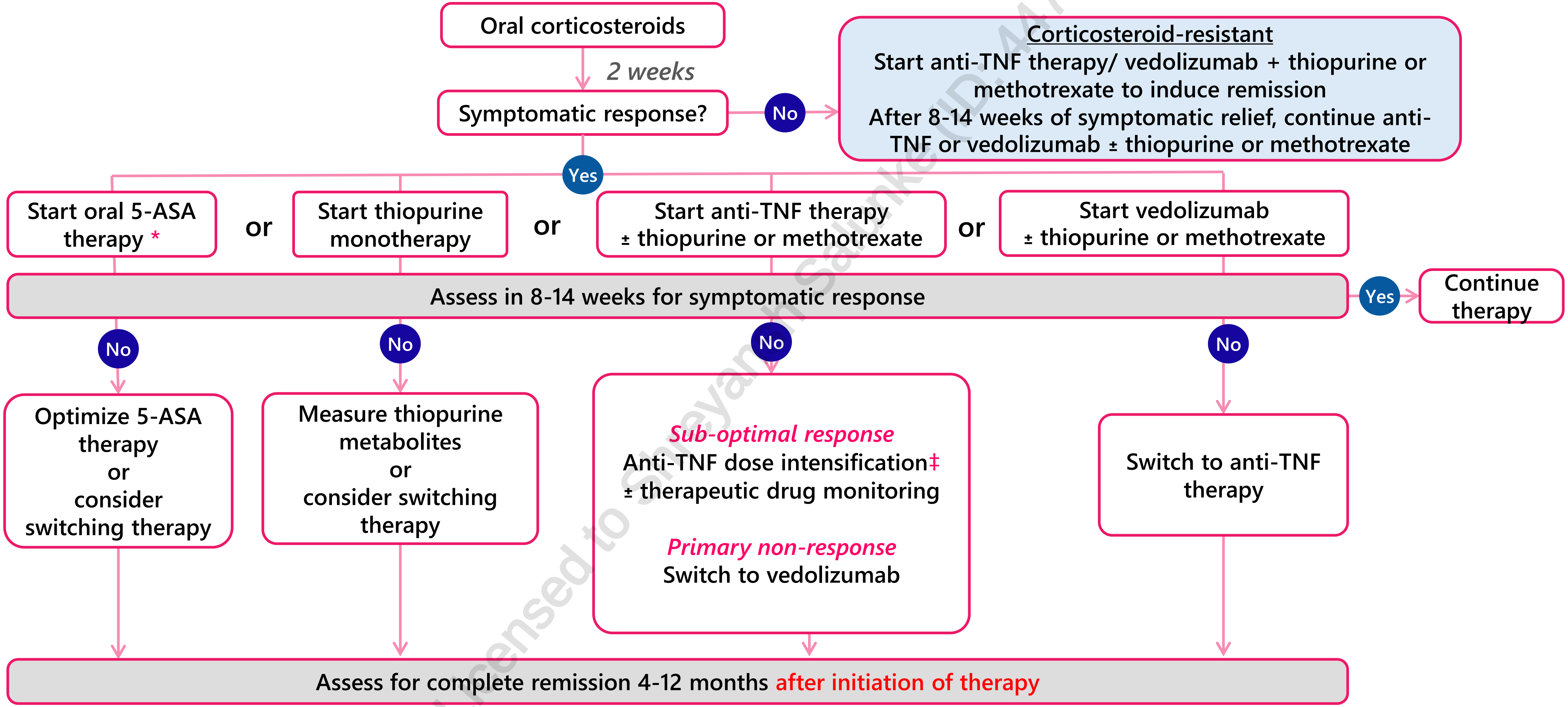
See moderate-to-severe algorithm to see how to proceed

* Combine oral + rectal 5-ASA if disease extent is beyond proctitis

† Do not switch routes

‡ Corticosteroids should only be used to induce remission

TREATMENT ALGORITHM- MODERATE-SEVERE UC



* Do not use if failed to respond to 5-ASA to induce remission in mild-to-moderate UC

NOTE: The role of dose intensification and therapeutic drug monitoring with vedolizumab therapy is uncertain

‡ (increase dose or shortening interval between doses)

TREATMENT ALGORITHM- MILD-MODERATE CD

Mild-to-moderate

- **Colon only***
  - *Induce remission with*
    sulfasalazine 4-6 g/day (mild disease)
    OR
    prednisone 40-60 mg/day
- **Terminal ileum ± right colon***
  - *Induce remission with*
    Oral Budesonide 9 mg/day

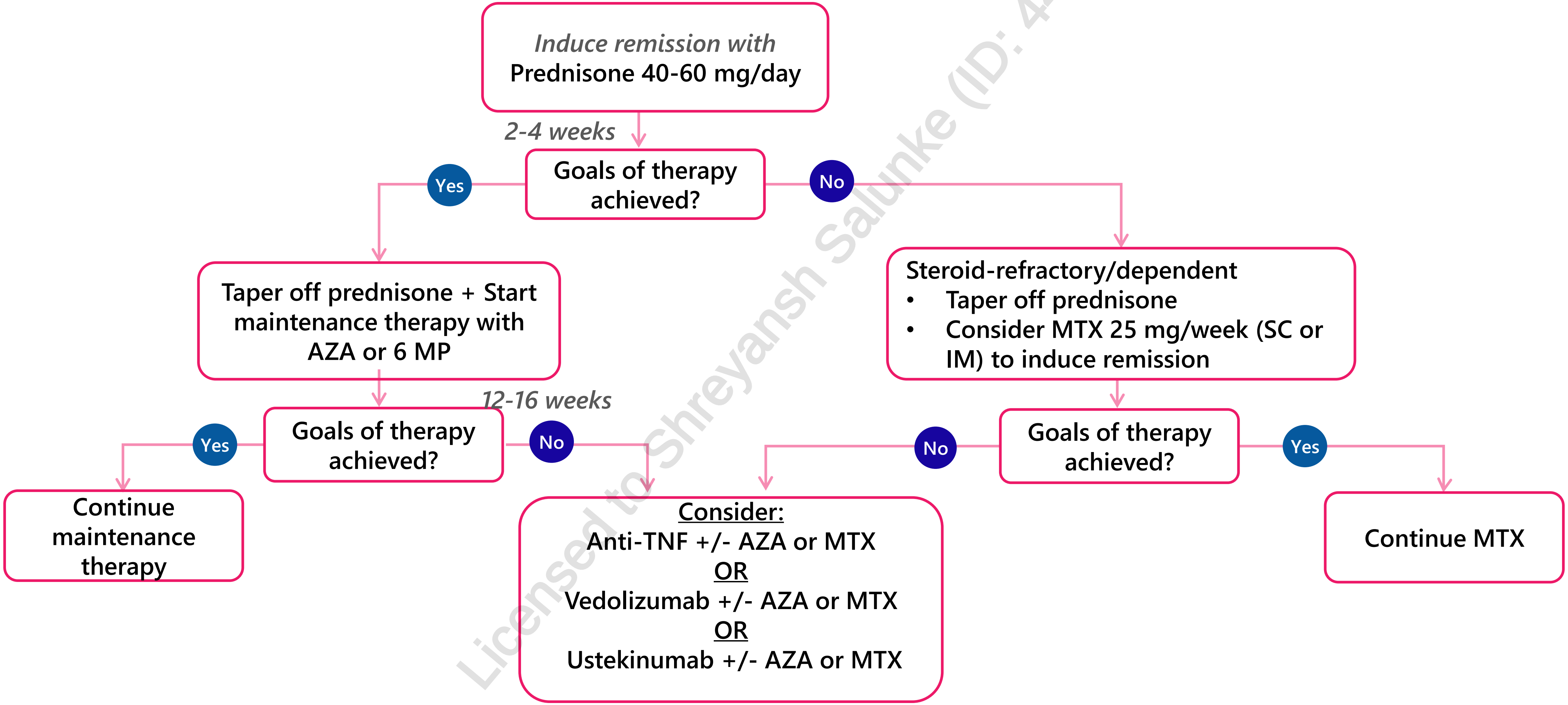
Sulfa: 2-4 months
Pred: 2-4 weeks
Budesonide: 4-8 weeks

Goals of therapy achieved?

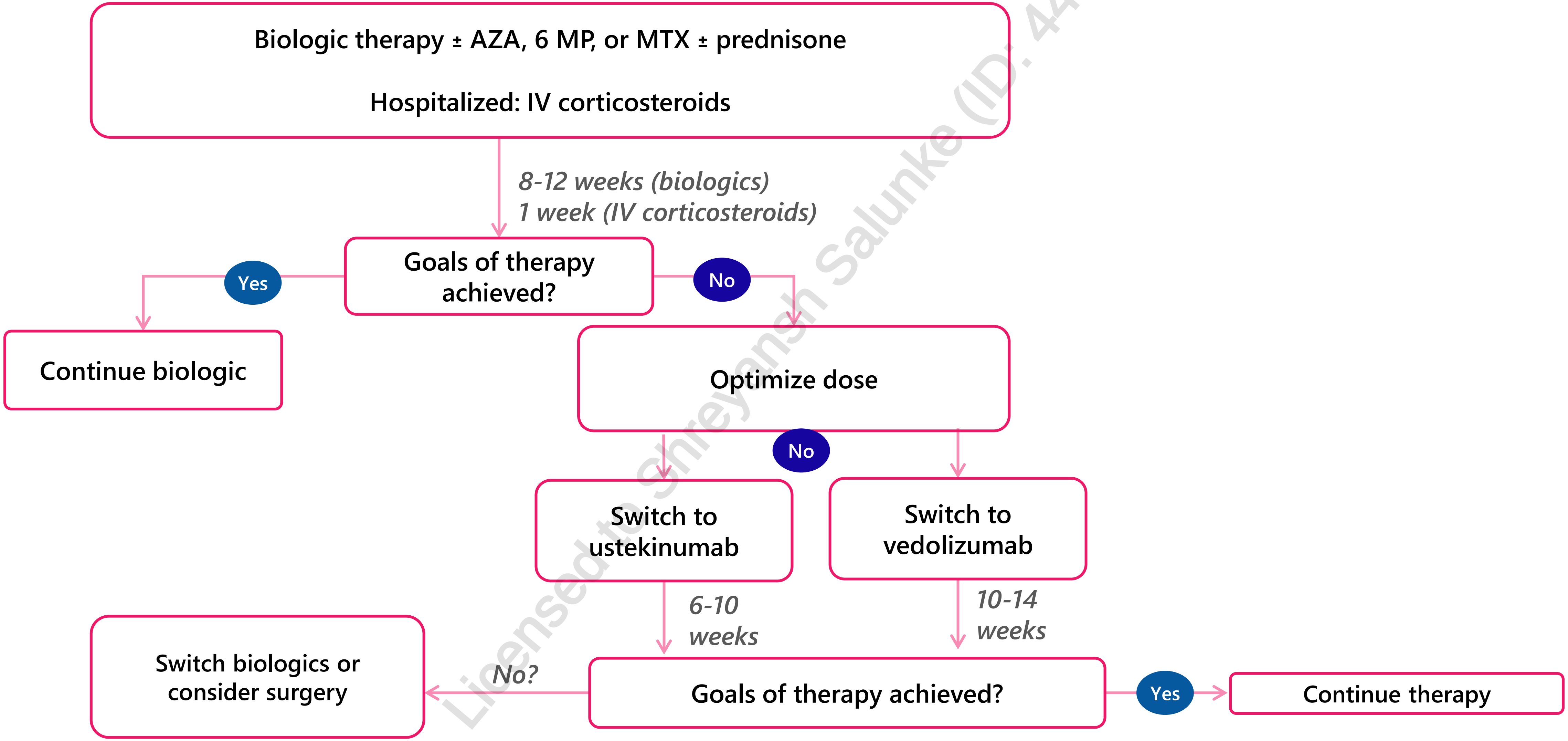
- **Yes**: Taper off prednisone or budesonide
  Consider maintenance therapy with AZA or 6 MP or sulfasalazine? (if worked initially)
- **No**: *Budesonide:* switch to prednisone 40-60mg/day to induce remission
  Consider treatment as low-risk, moderate to severe CD

*Consider using moderate-severe algorithm if patient has risk factors for poor prognosis

TREATMENT ALGORITHM- LOW RISK, MODERATE-SEVERE CD



TREATMENT ALGORITHM- HIGH RISK, MODERATE-SEVERE CD



MONITORING PARAMETERS

Pharmacotherapeutic endpoints		
Parameter	Degree of Change	Timeframe
Stool frequency	Improve Normalize (< 4x / day)	Weeks
Cramps, erythema, weight loss	Normalize	Weeks
Recurrent episodes	Prevent	Duration of therapy (onset of immunologics may take 3 months)

Safety endpoints		
Parameter	Degree of Change	Timeframe
ADRs of medications (headache, bone marrow suppression, etc.)	Prevent or minimize	Duration of therapy

TAKEAWAY

- CD can be anywhere from mouth to anus and is transmural
- UC is found in colon/rectum and affects mucosal layer
- Immunomodulators and biologics take months to work → use steroids to help induce a faster remission in moderate-severe disease
  - Taper off steroids slowly once remission is achieved
- Review of immunization status is essential prior to starting biologic therapy
- Review algorithms, slides 23-27 above

CASE

ST is a 24-year-old male of Afro-Caribbean descent, who presents with worsening abdominal pain and diarrhea in the past 2 months. ST currently has 5 bowel movements/day with some blood, increased from 3-4 BMs in the past. He has a low-grade fever, and abdominal pain that he has difficulty managing. He can tolerate the usual foods he likes, but he has a reduced appetite and poor energy levels. He has not had significant weight loss. He recently finished university and is now starting to deal with the stress of finding a job to pay off his school debt. ST smokes socially and drinks 6-7 drinks/week. He takes ibuprofen 400 mg po TID PRN for abdominal pain. He has no allergies to medications. His grandmother had a late diagnosis of Crohn's Disease at age 65, shortly after her husband of 50 years passed away.

Objective findings:

- WBC=12.1 (4.5-11.0 x10^9/L)
- Hgb=111 (140-180 g/L)
- SCr=78 (62-115 umol/L)
- T= 37.7° C
- Electrolytes normal
- Stool culture negative

CASE- QUESTIONS

1. All of the following risk factors may contribute to ST's potential IBD diagnosis EXCEPT?

   a) Family history
   b) Stress
   c) Age
   d) Race

2. ST is seen by a gastroenterologist and his scope shows mucosal erosions in the distal colon and rectum. Which diagnosis is most appropriate?

   a) Mild Crohn's Disease
   b) Colon cancer
   c) Moderate Crohn's Disease
   d) Moderate-severe ulcerative colitis

CASE- QUESTIONS

3. What treatment option would be most suitable for ST?

   a) Prednisone
   b) Oral +/- rectal 5-ASA
   c) Azathioprine
   d) Methotrexate

4. About 2 weeks later, the gastroenterologist deems that the therapy chosen above has not resulted in an improvement. All of the following are reasonable options to start EXCEPT:

   a) Infliximab
   b) Vedolizumab
   c) Infliximab + Azathioprine + Methotrexate
   d) Infliximab + Azathioprine

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CHANGE LOG

August 2025

- Slide 6: Moved the submucosa, muscularis + serosa arrows to right side to make diagram clearer

September 2025

- Slide 10: Removed ‘FYI’

November 2025

- Formatting edits made, no content changes

April 2026

- Slide 2: Updated NAPRA competencies
- Slide 11: Minor rephrasing
- Slide 31: Changed Question 1 answer choice C) to age

April 2026

- Slide 24: Updated corticosteroid-resistant algorithm section